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$$z^4 + 3z^2 + 2$$

$$\text{Stel } u = z^2$$

$$\Rightarrow u^2 + 3u + 2$$

$$D = 9 - 8 = 1$$

$$u = \frac{-3 \pm 1}{2}$$

$$u_1 = -1, u_2 = -2$$

$$\text{Voor } u = -1: z^2 = -1 \Rightarrow z = \pm i$$

$$\text{Voor } u = -2: z^2 = -2 \Rightarrow z = \pm i\sqrt{2}$$

$$\Rightarrow z^4 + 3z^2 + 2 = (z - i)(z + i)(z - i\sqrt{2})(z + i\sqrt{2})$$

$$z^4 + 3z^2 + 2 = (z^2 + 1)(z^2 + 2)$$

References